Symbolic Projection Notes for $B_{c1} \to B_c^{(*)}\gamma$

Working notes

July 12, 2026

1 Scope

These notes collect the compact FeynCalc projection results for the heavy-heavy channels

$$B_{c1}(6743) \to B_c\gamma, \qquad B_{c1}(6743) \to B_c^*\gamma, \qquad B_{c1}(6750) \to B_c\gamma, \qquad B_{c1}(6750) \to B_c^*\gamma.$$

The calculation is structurally different from the D_s and B_s light-cone sum rules. In the B_c system both valence constituents are heavy, so the leading radiative OPE is a hard-photon heavy-heavy calculation. There is no light spectator line to be replaced by photon distribution amplitudes, and no $\chi_s\langle\bar{s}s\rangle$ type term.

2 Currents and Propagators

For $B_c^+ \sim c\bar{b}$, we use

$$j_5 = \bar{b}\, i\gamma_5 c, \qquad j_\rho^V = \bar{b}\,\gamma_\rho c,$$

and the two axial basis currents

$$J_\mu^A = \bar{b}\,\gamma_\mu\gamma_5 c, \qquad J_\mu^B = \frac{i}{m_b + m_c}\,\bar{b}\,\sigma_{\mu\alpha}P^\alpha\gamma_5 c, \qquad P = p + q.$$

The physical currents are

$$J_{1\mu} = \sin\theta\, J_\mu^A + \cos\theta\, J_\mu^B, \qquad J_{2\mu} = \cos\theta\, J_\mu^A - \sin\theta\, J_\mu^B.$$

This mixing is used at the invariant-amplitude level. The OPE is first derived for the basis currents because the two Dirac traces are independent; the physical amplitudes are then

$$\widehat{\Pi}_1 = \sin\theta\,\widehat{\Pi}_A + \cos\theta\,\widehat{\Pi}_B, \qquad \widehat{\Pi}_2 = \cos\theta\,\widehat{\Pi}_A - \sin\theta\,\widehat{\Pi}_B.$$

For the pseudoscalar final state,

$$g_i = \frac{m_b + m_c}{m_{B_c}^2 f_{B_c}\, m_i f_i} \exp\left(\frac{m_i^2}{M_1^2} + \frac{m_{B_c}^2}{M_2^2}\right)\widehat{\Pi}_i^{(P)},$$

while for the vector final state,

$$g_i^* = \frac{1}{m_{B_c^*} f_{B_c^*}\, m_i f_i} \exp\left(\frac{m_i^2}{M_1^2} + \frac{m_{B_c^*}^2}{M_2^2}\right)\widehat{\Pi}_i^{(V)}.$$

The updated $B_c\gamma$ script applies this rotation with physical two-point residues f_1 and f_2. The vector $B_c^*\gamma$ entries remain diagnostic until their tensor-basis normalization is audited.

The free heavy-quark propagator in momentum space is

$$S_Q^{(0)}(k) = \frac{i(\not{k} + m_Q)}{k^2 - m_Q^2 + i0}, \qquad Q = c, b.$$

The hard photon is included by inserting the electromagnetic vertex on either heavy line:

$$S_Q^{(\gamma)}(k,k+q)=S_Q^{(0)}(k+q)\,ie_Q\gamma_\nu\epsilon^\nu\,S_Q^{(0)}(k).$$

For the $\bar{b}$ line it is convenient to use the effective antiquark charge $e_{\bar{b}}=+e/3$, while $e_c=+2e/3$. Equivalently, one may keep $e_b=-e/3$ and track the orientation of the antiquark line. The scripts use $e_{\bar{b}}$ explicitly.

Thus, yes: this is the same heavy-heavy propagator logic as in a B_c-mixing sum-rule calculation. The important difference from the strange-sector radiative calculation is that we do not insert a light-quark photon DA. If we later include power corrections, the natural heavy-heavy correction is the background-gluon expansion of the heavy propagators, e.g. gluon-condensate terms, not strange-quark condensate photon DAs.

The submitted B_c-mixing work uses the same local basis,

$$J_\mu^A=\bar{b}\gamma_\mu\gamma_5 c,\qquad J_\mu^B=\frac{i}{m_b+m_c}\bar{b}\sigma_{\mu\alpha}p^\alpha\gamma_5 c,$$

with $m_b=4.18$ GeV, $m_c=1.27$ GeV, and the final working window

$$7\ \text{GeV}^2\le M^2\le 9\ \text{GeV}^2,\qquad 53\ \text{GeV}^2\le s_0\le 55\ \text{GeV}^2.$$

Those are useful starting values for the heavy-heavy OPE in the radiative problem. They are not yet automatically the final radiative-transition windows; the final windows must be fixed by the three-point sum-rule stability, pole dominance, and convergence tests.

3 Step-by-Step Sum-Rule Construction

This section is written as a checklist for reproducing the calculation.

1. Start from basis currents. The physical currents J_1,J_2 are mixed combinations of J_A,J_B. We derive the OPE for J_A and J_B separately only because the Dirac traces are simpler. The physical amplitudes are obtained afterward through the $\sin\theta,\cos\theta$ rotation.

2. Write the correlators. For $B_c\gamma$,

$$\Pi_\mu^{(5,K)}(p,q)=i\int d^4x\,e^{ipx}\langle\gamma(q,\epsilon)|T\{\bar{b}(x)i\gamma_5c(x)\,J_{K\mu}^\dagger(0)\}|0\rangle,\qquad K=A,B.$$

For $B_c^*\gamma$,

$$\Pi_{\mu\nu}^{(V,K)}(p,q)=i\int d^4x\,e^{ipx}\langle\gamma(q,\epsilon)|T\{\bar{b}(x)\gamma_\nu c(x)\,J_{K\mu}^\dagger(0)\}|0\rangle.$$

3. Contract the heavy fields. There are two hard-photon diagrams: photon emission from the c line and photon emission from the $\bar{b}$ line. The free heavy propagator is

$$S_Q^{(0)}(k)=\frac{i(\not{k}+m_Q)}{k^2-m_Q^2+i0},$$

and the photon insertion is

$$S_Q^{(\gamma)}(k,k+q)=S_Q^{(0)}(k+q)\,ie_Q\not{\epsilon}\,S_Q^{(0)}(k).$$

The charge convention in the scripts is $e_c=+2/3$ and $e_{\bar{b}}=+1/3$. No photon DA or light-quark condensate term appears in this heavy-heavy leading baseline.

4. Evaluate and project the traces. For $B_c\gamma$ project onto

$$S_{\mu\nu} = p_\nu q_\mu - (p\cdot q)g_{\mu\nu}, \qquad \mathcal{P}^{\mu\nu} = \frac{S^{\mu\nu}}{2(p\cdot q)^2}, \qquad S_{\mu\nu}\mathcal{P}^{\mu\nu} = 1.$$

For $B_c^*\gamma$ the current diagnostic tensor is

$$V_{\mu\nu\rho} = p_\nu \epsilon_{\mu\rho p q} - (p\cdot q)\epsilon_{\mu\nu\rho p}, \qquad V_{\mu\nu\rho}V^{\mu\nu\rho} = -4p^2(p\cdot q)^2.$$

The vector-channel result still needs matching to the physical $1^+ \to 1^-\gamma$ amplitude basis.

5. Reduce to denominator structures. For photon emission from the charm line use

$$D_1 = (k+q)^2 - m_c^2, \quad D_2 = k^2 - m_c^2, \quad D_3 = (k-p)^2 - m_b^2.$$

For photon emission from the anti-bottom line use

$$D_1 = k^2 - m_c^2, \quad D_2 = (k-p)^2 - m_b^2, \quad D_3 = (k-p+q)^2 - m_b^2.$$

The triangle pieces contain the full three-denominator product. Terms where a numerator cancels one denominator become two-denominator/contact terms and must be included or shown to vanish.

6. Borel match and rotate. The true variables are p^2 and $P^2 = (p+q)^2$, with

$$p\cdot q = \frac{P^2 - p^2}{2}.$$

The pilot numerical calculation uses the diagonal choice $M_1^2 = M_2^2 = 2M^2$, leading to the practical exponential

$$\exp\left[\frac{m_i^2 + m_f^2}{2M^2}\right].$$

After obtaining $\widehat{\Pi}_A, \widehat{\Pi}_B$, the physical combinations are

$$\widehat{\Pi}_{6743} = \sin\theta\,\widehat{\Pi}_A + \cos\theta\,\widehat{\Pi}_B, \qquad \widehat{\Pi}_{6750} = \cos\theta\,\widehat{\Pi}_A - \sin\theta\,\widehat{\Pi}_B.$$

For the $B_c\gamma$ widths we use

$$\Gamma(1^+ \to 0^-\gamma) = \frac{\alpha_{\rm em}}{3} g^2 E_\gamma^3, \qquad E_\gamma = \frac{m_i^2 - m_f^2}{2m_i}.$$

4 $B_{c1} \to B_c\gamma$: E1 Projection

For the pseudoscalar final state we project onto

$$S_{\mu\nu} = p_\nu q_\mu - (p\cdot q)g_{\mu\nu}, \qquad \mathcal{P}_{\mu\nu} = \frac{S_{\mu\nu}}{2(p\cdot q)^2}.$$

FeynCalc gives the normalization check

$$S_{\mu\nu}\mathcal{P}^{\mu\nu} = 1.$$

We denote

$$p^2 = p_2, \qquad p\cdot q = pq.$$

After rewriting loop scalar products in terms of inverse denominators and setting those denominators to zero, the projected triangle-core numerators are

$$
\begin{aligned}
C_A^{(c)} &= -4i\, m_c + \frac{4i\, m_b m_c^2}{pq} - \frac{4i\, m_c^3}{pq}, \\
C_A^{(\bar{b})} &= -4i\, m_b + \frac{4i\, m_b^3}{pq} - \frac{4i\, m_b^2 m_c}{pq},
\end{aligned}
$$

so that

$$C_A = e_c C_A^{(c)} + e_{\bar{b}} C_A^{(\bar{b})}.$$

For the tensor current,

$$
\begin{aligned}
C_B^{(c)} &= \frac{4i\, m_b m_c - 8i\, m_c^2}{m_b + m_c} - \frac{4i\, m_c^2 p_2}{(m_b + m_c) pq}, \\
C_B^{(\bar{b})} &= -\frac{4i\, m_b m_c}{m_b + m_c} - \frac{4i\, m_b^2 p_2}{(m_b + m_c) pq},
\end{aligned}
$$

and

$$C_B = e_c C_B^{(c)} + e_{\bar{b}} C_B^{(\bar{b})}.$$

The denominator-cancellation terms are stored in the corresponding output file. They should not be discarded automatically; they correspond to reduced two-denominator or contact contributions and must be treated in the dispersion/Borel analysis.

5 $B_{c1} \to B_c^* \gamma$: Levi-Civita Projection

For the vector final state a useful gauge-invariant structure is

$$V_{\mu\nu\rho} = p_\nu\, \epsilon_{\mu\rho p q} - (p \cdot q)\, \epsilon_{\mu\nu\rho p}.$$

It is transverse in the photon index for $q^2 = 0$. The FeynCalc checks are

$$V_{\mu\nu\rho} V^{\mu\nu\rho} = -4 p_2 (pq)^2, \qquad V_{\mu\nu\rho} \mathcal{P}^{\mu\nu\rho} = 1.$$

The projected J_A triangle cores are

$$
\begin{aligned}
C_{A,V}^{(c)} &= -\frac{4i\, m_b m_c}{p_2} + \frac{2i\, m_c^2}{p_2} + \frac{4i\, m_c^2}{pq}, \\
C_{A,V}^{(\bar{b})} &= -\frac{2i\, m_b^2}{p_2} + \frac{4i\, m_b m_c}{p_2} + \frac{4i\, m_b^2}{pq}.
\end{aligned}
$$

The tensor-current vector cores are

$$
\begin{aligned}
C_{B,V}^{(c)} &= \frac{2i\, m_c}{m_b + m_c} + \frac{2i\, m_b^2 m_c - 4i\, m_b m_c^2 + 2i\, m_c^3}{(m_b + m_c) p_2} \\
&\quad - \frac{4i\, m_b m_c^2 - 4i\, m_c^3}{(m_b + m_c) pq} + \frac{2i\, m_c\, pq}{(m_b + m_c) p_2}, \\
C_{B,V}^{(\bar{b})} &= \frac{2i\, m_b}{m_b + m_c} + \frac{-6i\, m_b^3 + 4i\, m_b^2 m_c + 2i\, m_b m_c^2}{(m_b + m_c) p_2} \\
&\quad - \frac{4i\, m_b^3 - 4i\, m_b^2 m_c}{(m_b + m_c) pq} + \frac{2i\, m_b\, pq}{(m_b + m_c) p_2}.
\end{aligned}
$$

Again,

$$C_{A,V} = e_c C_{A,V}^{(c)} + e_{\bar{b}} C_{A,V}^{(\bar{b})}, \qquad C_{B,V} = e_c C_{B,V}^{(c)} + e_{\bar{b}} C_{B,V}^{(\bar{b})}.$$

6 Controlled $B_c\gamma$ Pass

The script `Bc_gamma/scripts/bc_ps_complete_analysis.py` uses the compact $B_c\gamma$ triangle cores above and replaces the earlier common axial-vector normalization by physical two-point residues. We use the same central heavy-quark inputs and Borel window as the submitted B_c-mixing analysis,

$$m_b = 4.18 \text{ GeV}, \qquad m_c = 1.27 \text{ GeV}, \qquad \theta = 43.3^\circ,$$

more precisely $\theta_{\text{mean}} = 43.295^\circ$ with $\sigma_\theta = 0.151^\circ$ and 16–84% interval $[43.147^\circ, 43.455^\circ]$ from the B_c-mixing Monte Carlo. The updated Monte Carlo samples this θ distribution.

$$M^2 = \{7, 8, 9\} \text{ GeV}^2, \qquad s_0 = \{53, 54, 55\} \text{ GeV}^2.$$

The pseudoscalar decay-constant benchmark is

$$f_{B_c} = 0.371 \pm 0.037 \text{ GeV}.$$

The physical axial-vector residues are extracted from the same two-point B_c-mixing basis. With $s_\theta = \sin\theta$ and $c_\theta = \cos\theta$,

$$\Pi_1 = s_\theta^2 \Pi_{AA} + 2 s_\theta c_\theta \Pi_{AB} + c_\theta^2 \Pi_{BB}, \qquad \Pi_2 = c_\theta^2 \Pi_{AA} - 2 s_\theta c_\theta \Pi_{AB} + s_\theta^2 \Pi_{BB},$$

and

$$f_i(M^2, s_0) = \left[\frac{e^{m_i^2/M^2} \Pi_i(M^2, s_0)}{m_i^2} \right]^{1/2}.$$

Using the perturbative heavy-heavy two-point spectral densities over the window above gives

$$f_1 = 0.264^{+0.005}_{-0.005} \text{ GeV}, \qquad f_2 = 0.722^{+0.016}_{-0.013} \text{ GeV}$$

from the 16–84% grid interval. This replaces the older common benchmark $f_{B_{c1}} = 0.373$ GeV.

Table 1: Updated $B_c\gamma$ results with physical two-point residues. The quoted intervals are the Monte Carlo 16–84% ranges over $m_b, m_c, M^2, s_0, \theta, f_{B_c}$, with f_1, f_2 recomputed from the two-point perturbative moments.

Channel	Status	Coupling/projection coefficient	Γ (keV)
$B_{c1}(6743) \to B_c\gamma$	controlled $B_c\gamma$	$+0.0315\,[0.0269, 0.0370]$ GeV^{-1}	$0.223\,[0.162, 0.307]$
$B_{c1}(6750) \to B_c\gamma$	controlled $B_c\gamma$	$-0.184\,[-0.205, -0.167]$ GeV^{-1}	$8.01\,[6.60, 9.90]$

For the vector final state we also remove the fixed-normalization assumption. The final-state vector decay constant is defined by

$$\langle 0 | \bar{c} \gamma_\rho b | B_c^*(p, \eta) \rangle = m_{B_c^*} f_{B_c^*} \eta_\rho.$$

Projecting the two-point vector-current correlator onto the spin–1 invariant gives the perturbative density

$$\rho_V(s) = \frac{3}{16\pi^2} \frac{\sqrt{\lambda(s, m_b^2, m_c^2)}}{s} \left[-\frac{8(k \cdot p)^2}{s} - 4m_c^2 + 12(k \cdot p) + 12 m_b m_c \right],$$

where $k \cdot p = (s + m_c^2 - m_b^2)/2$. In the equal-mass limit this reduces to the expected S-wave vector-current density proportional to $(s + 2m_Q^2)\sqrt{1 - 4m_Q^2/s}$. The corresponding two-point sum rule is

$$f_{B_c^*}^2 m_{B_c^*}^2 e^{-m_{B_c^*}^2/M_V^2} = \int_{(m_b+m_c)^2}^{s_{0,V}} ds\, e^{-s/M_V^2} \rho_V(s).$$

Using $M_V^2 = 5$–8 GeV^2 and $s_{0,V} = 40$–44 GeV^2, the direct spin-projector density gives

$$f_{B_c^*}^{\text{projector}} = 0.753\,[0.693, 0.808] \text{ GeV}.$$

This is a projector-convention diagnostic. For the physical matrix element $\langle 0|\bar{c}\gamma_\rho b|B_c^*\rangle = m_{B_c^*} f_{B_c^*} \eta_\rho$, the standard vector-invariant convention divides the density by the transverse spin-trace factor of three. The corresponding self-contained two-point result is

$$f_{B_c^*}^{\text{standard}} = 0.435\,[0.400, 0.467] \text{ GeV}.$$

This is close to the commonly used reference value $f_{B_c^*} = 0.384 \pm 0.038$ GeV. We therefore quote the standard normalization as the preferred self-contained result and keep the reference normalization as a sensitivity check:

Table 2: Leading hard-QCDSR $B_c^*\gamma$ vector baselines. These entries include the perturbative hard-photon three-point term and the physical f_1, f_2 residues. Radiative G^2 and vector contact terms are not included.

Channel	Normalization	Coupling/projection coefficient	Γ (keV)
$B_{c1}(6743) \to B_c^*\gamma$	standard vector-invariant $f_{B_c^*}$	$-0.0785\,[-0.0843, -0.0736]$ GeV^{-1}	$74.3\,[65.3, 85.5]$
$B_{c1}(6750) \to B_c^*\gamma$	standard vector-invariant $f_{B_c^*}$	$+0.168\,[0.156, 0.180]$ GeV^{-1}	$360\,[312, 412]$
$B_{c1}(6743) \to B_c^*\gamma$	reference $f_{B_c^*} = 0.384 \pm 0.038$ GeV	$-0.0877\,[-0.0986, -0.0795]$ GeV^{-1}	$92.6\,[76.1, 117]$
$B_{c1}(6750) \to B_c^*\gamma$	reference $f_{B_c^*} = 0.384 \pm 0.038$ GeV	$+0.187\,[0.169, 0.210]$ GeV^{-1}	$447\,[365, 564]$

The $B_c\gamma$ entries are the controlled hard-photon baseline from the present calculation. The vector entries are no longer just placeholders: they use the $V_{\mu\nu\rho}$ projection, the physical f_1, f_2 residues, and an explicit $f_{B_c^*}$ normalization. The projector diagnostic gives $24.8[21.8, 28.5]$ and $120[104, 137]$ keV for the two vector channels and is kept only in the output CSV files as a convention check. The table above uses the standard vector-current convention. These entries should nevertheless be described as leading perturbative vector baselines, not full-OPE final predictions, until the radiative G^2 and contact sectors are completed.

The vector stability plots are generated by `bc_vec_complete_analysis.py`. On the central 3×3 (M^2, s_0) grid, the standard-normalization widths vary from about 66.7 to 80.9 keV for $B_{c1}(6743) \to B_c^*\gamma$, and from about 310 to 406 keV for $B_{c1}(6750) \to B_c^*\gamma$. The large higher state width is therefore not a numerical instability of the selected Borel window.

For the missing radiative G^2/contact sector we use the same conservative screening logic as in the $B_c\gamma$ channel, but with one additional residue factor. The vector sum rule contains $f_i f_{B_c^*}$ in the denominator, so

$$\left|\frac{\delta g}{g}\right| \lesssim \left|\frac{\delta\Pi^{(3)}}{\Pi^{(3)}}\right| + \left|\frac{\delta f_i}{f_i}\right| + \left|\frac{\delta f_{B_c^*}}{f_{B_c^*}}\right| \simeq 2 \max \left|\frac{\Pi_{G^2}}{\Pi_{\text{pert}}}\right|,$$

and

$$\left|\frac{\delta\Gamma}{\Gamma}\right| \lesssim 4 \max \left|\frac{\Pi_{G^2}}{\Pi_{\text{pert}}}\right|.$$

For the preferred standard vector normalization this gives the screening envelopes

channel	median envelope	maximum envelope
$B_{c1}(6743) \to B_c^*\gamma$	± 11.1 keV	± 27.7 keV
$B_{c1}(6750) \to B_c^*\gamma$	± 53.6 keV	± 139 keV

for the widths. This is not an explicit condensate correction; it is a publication-level warning that the vector-channel result should be called a leading perturbative baseline until the all-positive radiative G^2 sector is integrated explicitly. The refined contact audit below shows that the contact rows do not have ordinary physical two-channel support; the crossed rows are retained only as an audit trail.

7 Comparison with Experiment and Other Theory

LHCb has observed a wide peaking structure in the $B_c^+\gamma$ mass spectrum with significance above seven standard deviations. A minimal two-peak description gives [1, 2]

$$M_1 = 6704.8 \pm 5.5 \pm 2.8 \pm 0.3 \text{ MeV}, \qquad M_2 = 6752.4 \pm 9.5 \pm 3.1 \pm 0.3 \text{ MeV},$$

where the errors are statistical, systematic, and due to the B_c^+ mass. No individual radiative partial width is measured. The natural widths are expected to be only a few hundred keV and are neglected in the experimental fit because they are small compared with the photon-energy resolution.

Table 3: Qualitative comparison with the current experimental information from LHCb [1, 2].

Quantity	Experiment	Present input/result
Observed $B_c(1P)^+$-like signal	yes, $> 7\sigma$ in $B_c^+\gamma$	channels studied explicitly
Lower visible peak	$6704.8 \pm 5.5 \pm 2.8 \pm 0.3$ MeV	input state 6743 MeV
Higher visible peak	$6752.4 \pm 9.5 \pm 3.1 \pm 0.3$ MeV	input state 6750 MeV
Partial widths	not measured	$B_c\gamma$: 0.223, 8.01 keV

The lower visible peak should not be identified one-to-one with the input 6743 MeV state without care. In the measured $B_c\gamma$ spectrum, decays through $B_c^*\gamma$, followed by an unreconstructed soft $B_c^* \to B_c\gamma$, can shift visible peaks by the unknown hyperfine splitting.

For comparison with other theory we should not compress different papers into a single range. Table 4 lists each quoted calculation separately. The entries are the values collected in Bondar–Milstein [4]; the original model papers are cited in the first column.

Table 4: Comparison with representative predictions for $B_{c1} \to B_c^{(*)}\gamma$. Widths are in keV.

Publication	$6743 \to B_c\gamma$	$6743 \to B_c^*\gamma$	$6750 \to B_c\gamma$	$6750 \to B_c^*\gamma$
Godfrey [5]	13	60	80	11
Ebert–Faustov–Galkin [6]	18.4	78.9	132	13.6
Fulcher [7]	32.4	75.6	127.8	26.2
Gershtein et al. [8]	11.6	77.8	131.1	8.1
T. Y. Li et al. [9]	16.6	49	66.6	10.5
Q. Li et al. [10]	35	70	74	40
X. J. Li et al. [11]	30.1	47.8	64	25.6
Eichten–Quigg [12]	9.9	62.5	92.3	7.5
Bondar–Milstein [4]	22	75	47	2
This work	$0.223[0.162, 0.307]$	$46.3[41.0, 53.8]$	$8.01[6.60, 9.90]$	$1.67[1.43, 2.04] \times 10^3$

Thus the $B_c\gamma$ baseline is already useful for comparison, while the $B_c^*\gamma$ channel must still be completed with the physical tensor-basis normalization and the full double-Borel/contact-term treatment.

8 Perturbative and Condensate Content

The $B_c\gamma$ numerical table above contains the leading hard-photon perturbative three-point baseline, normalized with perturbative two-point physical residues. In practical terms this means:

- the photon is emitted from the charm or anti-bottom heavy-quark line;
- both heavy propagators are free propagators in the radiative three-point OPE;
- the retained spectral density is the perturbative triangle hard contribution generated by the projected Dirac traces;
- no photon DA, $\chi\langle\bar{q}q\rangle$, or light-quark condensate contribution appears, because there is no light valence quark in B_c;
- gluon-condensate corrections from the background-gluon expansion of the heavy propagators are not yet included in the radiative three-point correlator;
- reduced two-denominator/contact terms from denominator cancellation still require a full double-Borel audit, especially in the vector $B_c^*\gamma$ channel.

Thus the present $B_c\gamma$ result is best described as a controlled leading perturbative hard-QCDSR baseline. A final publication-level B_c analysis should add, or explicitly bound, the heavy-heavy gluon-condensate corrections to the three-point function. The $B_c^*\gamma$ channel also requires the contact-term and tensor-normalization audit discussed above.

9 Dimension-4 Radiative Gluon-Condensate Workbench

The heavy-heavy B_c system has no photon DA or light-quark condensate contribution, but the background-gluon expansion of the heavy propagators generates possible dimension-4 gluon-condensate corrections to the radiative three-point function. To keep the calculation reproducible, the $\langle\alpha_s G^2/\pi\rangle$ part was split into small FeynCalc chunks instead of reducing one very large trace.

For the $B_c\gamma$ E1 structure the two hard-photon topologies are

$$S_c(k+q)\gamma_{\rm em}S_c(k)\gamma_5 S_b(k-p), \qquad S_c(k)\gamma_5 S_b(k-p)\gamma_{\rm em}S_b(k-p+q).$$

In each topology we considered

- one single-line insertion $S_Q^{(G^2)}$ on each heavy propagator segment;
- one open-gluon-pair insertion $S^{(G)}S^{(G)}$ for each pair of propagator segments.

Thus there are six single-line topologies and six open-pair topologies. Each topology is evaluated for the two basis currents J_A and J_B, giving 24 projected numerator rows in total.

The projection uses

$$S_{\mu\nu} = p_\nu q_\mu - (p\cdot q)g_{\mu\nu}, \qquad \mathcal{P}^{\mu\nu} = \frac{S^{\mu\nu}}{2(p\cdot q)^2},$$

so that the projected coefficient is the coefficient of the same E1 tensor used in the perturbative baseline. With the present conventions the audit is

class	current-split rows	projected terms
$S_Q^{(G^2)}$	12	221
$S^{(G)}S^{(G)}$	12	164
total	24	385

The scripts which generate these checks are

- `Bc_gamma/scripts/step3_ps_g2_topology_inventory.py`;

- Bc_gamma/scripts/step3_ps_g2_single_line_cphoton.wl;
- Bc_gamma/scripts/step3_ps_g2_single_line_bbarphoton.wl;
- Bc_gamma/scripts/step3_ps_g2_open_pair_cphoton.wl;
- Bc_gamma/scripts/step3_ps_g2_open_pair_bbarphoton.wl;
- Bc_gamma/scripts/step3_ps_g2_projection_summary.py;
- Bc_gamma/scripts/step3_ps_g2_denominator_bookkeeping.py.

The raw outputs are stored in Bc_gamma/outputs/step3_ps_g2_*.txt and the summary table is stored in Bc_gamma/outputs/step3_ps_g2_projection_summary.csv. The denominator manifest is stored in Bc_gamma/outputs/step3_ps_g2_denominator_bookkeeping.csv.

The denominator bookkeeping is most compact in Schwinger parameters. For the charm-line photon topology we use

$$a \leftrightarrow S_c(k+q), \qquad b \leftrightarrow S_c(k), \qquad r \leftrightarrow S_b(k-p), \qquad A = a+b+r.$$

After shifting the loop momentum the quadratic form is

$$\Phi_c = \frac{r(a+b)}{A}p^2 + \frac{2ar}{A}(p\cdot q) - (a+b)m_c^2 - rm_b^2 = \frac{rb}{A}p^2 + \frac{ar}{A}P^2 - (a+b)m_c^2 - rm_b^2.$$

For the anti-bottom-line photon topology we use

$$a \leftrightarrow S_c(k), \qquad b \leftrightarrow S_b(k-p), \qquad r \leftrightarrow S_b(k-p+q), \qquad A = a+b+r,$$

which gives

$$\Phi_{\bar{b}} = \frac{a(b+r)}{A}p^2 + \frac{2r(b+r)}{A}(p\cdot q) - am_c^2 - (b+r)m_b^2.$$

Equivalently, since $P^2 = p^2 + 2p\cdot q$,

$$\Phi_{\bar{b}} = \frac{(b+r)(a-r)}{A}p^2 + \frac{r(b+r)}{A}P^2 - am_c^2 - (b+r)m_b^2.$$

For $S_Q^{(G^2)}$, the selected propagator power is four and the other two propagator powers are one. For $S^{(G)}S^{(G)}$, the selected pair carries powers two and two, while the remaining propagator has power one.

This completes the numerator-projection stage of the radiative gluon-condensate workbench and attaches denominator powers to each projected row. It is not yet a numerical condensate correction to the decay width. The next step is to perform the double-dispersion/double-Borel reduction and integrate the result over the same Borel and continuum-threshold windows used for the perturbative baseline.

Conservative G^2 Size Screen

Before completing the full derivative-delta reduction, we performed a conservative size screen using the completed B_c-mixing OPE convergence table in the same numerical window,

$$M^2 = \{7,8,9\}\,\mathrm{GeV}^2, \qquad s_0 = \{53,54,55\}\,\mathrm{GeV}^2.$$

The script is bc_ps_g2_screening_estimate.py. The corresponding CSV and text outputs are stored in the Bc_gamma/outputs directory.

The screen defines

$$\Delta_{2\mathrm{pt}}^{G^2} = \max\left(\left|\frac{\Pi_{AA}^{G^2}}{\Pi_{AA}^{\mathrm{pert}}}\right|, \left|\frac{\Pi_{AB}^{G^2}}{\Pi_{AB}^{\mathrm{pert}}}\right|, \left|\frac{\Pi_{BB}^{G^2}}{\Pi_{BB}^{\mathrm{pert}}}\right| \right)$$

from the B_c-mixing calculation. Since $f \sim \sqrt{\Pi}$, the normalization shift is estimated as $\Delta_f \simeq \Delta_{2\mathrm{pt}}^{G^2}/2$. Taking the unknown radiative three-point G^2 shift to be of the same size gives the conservative envelope

$$\left|\frac{\delta g}{g}\right|_{\mathrm{env}} = \Delta_{2\mathrm{pt}}^{G^2} + \frac{1}{2}\Delta_{2\mathrm{pt}}^{G^2}, \qquad \left|\frac{\delta \Gamma}{\Gamma}\right|_{\mathrm{env}} \simeq 2\left|\frac{\delta g}{g}\right|_{\mathrm{env}}.$$

Numerically this gives a median conservative envelope $|\delta g/g|_{\mathrm{env}} = 0.058$ and $|\delta\Gamma/\Gamma|_{\mathrm{env}} = 0.115$, with maxima 0.128 and 0.257, respectively, over the grid.

Therefore the screen is not small enough to justify silently dropping G^2 at publication level. It does not prove that the radiative three-point G^2 correction is as large as the envelope; it shows that a few-percent amplitude effect is plausible in the same heavy-heavy OPE window. The safe statement is that the present widths are leading perturbative hard-QCDSR baselines until the explicit radiative G^2 reduction is completed or bounded more sharply.

The channel-by-channel conservative envelope is summarized in Table 5. The table is generated by `bc_ps_g2_corrected_summary.py` and should be interpreted as a systematic bound, not as the final explicit radiative G^2 contribution.

Table 5: Perturbative $B_{c1} \to B_c\gamma$ result with conservative G^2 width envelope. Widths are in keV; the last column gives median/max envelope sizes.

Channel	$g_{\mathrm{pert}}\,[\mathrm{GeV}^{-1}]$	Γ_{pert}	G^2 env.
$B_{c1}(6743) \to B_c\gamma$	$0.0316[0.0269, 0.0370]$	$0.223[0.162, 0.308]$	$\pm 0.025/\pm 0.060$
$B_{c1}(6750) \to B_c\gamma$	$-0.184[-0.205, -0.167]$	$8.01[6.60, 9.90]$	$\pm 0.914/\pm 2.30$

Denominator-Reduced G^2 Inventory

To test whether the full explicit radiative G^2 correction could be obtained from ordinary positive-power Schwinger integrals alone, we reduced the projected G^2 numerators in inverse propagator denominators d_1, d_2, d_3. This is done by `step3_ps_g2_denominator_reduction.wl`; the summary is generated by `step3_ps_g2_denominator_reduction_summary.py`.

The complete reduction contains 747 terms. Of these, 375 have all effective denominator powers positive, while 372 terms have at least one nonpositive effective power and therefore belong to the contact/derivative sector. Thus the latter sector is not a small correction to the bookkeeping; it is roughly half of the denominator-reduced expression.

Consequently, an integration of only the all-positive sector would not be the full explicit radiative G^2 correction. The full result requires the derivative double-Borel treatment of the contact sector, analogous in spirit to the direct-Borel condensate machinery used in the B_c-mixing analysis. In the absence of that final derivative-contact reduction, the conservative envelope in Table 5 is the defensible numerical statement.

$B_c^*\gamma$ Denominator-Reduced G^2 Inventory

The same denominator-reduction audit was carried out for the vector final state. The projected structure is

$$V_{\mu\nu\rho} = p_\nu \epsilon_{\mu\rho p q} - (p\cdot q)\epsilon_{\mu\nu\rho p}, \qquad V_{\mu\nu\rho}V^{\mu\nu\rho} = -4p^2(p\cdot q)^2.$$

The script `step3_vec_g2_denominator_reduction.wl` replaces the appropriate propagator numerators by $S_Q^{(G^2)}$ or by open one-gluon pieces $S_Q^{(G)} S_{Q'}^{(G)}$, projects onto $V_{\mu\nu\rho}$, and rewrites the result in powers of d_1, d_2, d_3. The summary script is `step3_vec_g2_denominator_reduction_summary.py`.

The complete vector reduction contains 1095 terms:

sector	number of terms
all-positive denominator powers	538
contact/derivative sector	557

By current, the split is

current	all-positive	contact/derivative
J_μ^A	200	148
J_μ^B	338	409

and by background-field class it is

class	all-positive	contact/derivative
single-line G^2	209	368
open-pair GG	329	189

.

Thus the contact/derivative sector is slightly larger than the ordinary positive-denominator sector in the vector channel. This is why the $B_c^*\gamma$ result should be quoted as a leading perturbative vector baseline with a G^2/contact systematic screen, rather than as a completed full-OPE prediction.

The support-level double-Borel audit is performed by `step3_vec_g2_contact_support_audit.py`. It first separates contact rows that contain both hadronic virtualities from rows that are single-virtuality or no-external-virtuality subtraction terms. The first-pass result is

contact support class	number of rows
double-Borel-active candidates	131
single-virtuality rows	309
no-external-virtuality rows	117

All 131 first-pass active candidates come from the anti-bottom photon topology with remaining denominator support $\{d_1, d_3\}$. This needs one more physical-support check. In that topology

$$d_1 = k^2 - m_c^2, \qquad d_3 = (k - p + q)^2 - m_b^2.$$

Combining only d_1 and d_3 with Schwinger parameters gives an external piece proportional to

$$\frac{t_1 t_3}{t_1 + t_3}\left[-P^2 + 2p^2\right] = \frac{t_1 t_3}{t_1 + t_3}(p - q)^2, \qquad P = p + q, \quad q^2 = 0.$$

Thus the two-denominator contact candidate depends on the crossed virtuality $(p-q)^2$, not on independent physical initial- and final-channel pole variables. The refined audit `step3_vec_g2_contact_borel_admissibility.py` gives

refined contact support class	number of rows
crossed single-combination rows	131
single-virtuality rows	309
no-external-virtuality rows	117
ordinary two-channel contact rows	0

Therefore no contact/derivative row is added directly to the physical double-pole numerical sum in the standard two-channel Borel projection. The 131 crossed rows are stored in `step3_vec_g2_contact_active_candidates.csv` as an audit trail; they would require a separate crossed-contact prescription before being used in a numerical width.

For a student checking this calculation, the most useful debugging sequence is:

1. verify the projector normalization $V_{\mu\nu\rho}V^{\mu\nu\rho} = -4p^2(p \cdot q)^2$;
2. reproduce one J_μ^A single-line insertion before running all topologies;
3. check the denominator substitutions separately for charm-photon and anti-bottom-photon emission;
4. classify terms with effective powers $n_i > 0$ as ordinary Schwinger terms and terms with at least one $n_i \leq 0$ as contact/derivative terms;
5. run the support audit, then the refined Borel-admissibility audit;
6. check that the apparent 131 active rows all reduce to crossed $(p-q)^2$ support;
7. keep these rows in the audit trail, but do not add them to the physical double-pole sum unless a separate crossed-contact prescription is introduced.

10 Reproduction Checklist for the Present Pilot

The current numbers can be reproduced in the following order.

1. Run `Bc_gamma/scripts/step1_hard_photon_numerators.wl`. This computes the raw FeynCalc traces for photon emission from the charm line and from the anti-bottom line.
2. Run `Bc_gamma/scripts/step2_ps_e1_triangle_reduction.wl`. This projects the $B_c\gamma$ traces onto $S_{\mu\nu} = p_\nu q_\mu - (p \cdot q)g_{\mu\nu}$ and checks $S_{\mu\nu}\mathcal{P}^{\mu\nu} = 1$.
3. Run `Bc_gamma/scripts/step2_vec_epsilon_A_reduction.wl` and `Bc_gamma/scripts/step2_vec_epsilon_B_reduction.wl`. These produce the diagnostic $B_c^*\gamma$ cores and check the normalization of $V_{\mu\nu\rho}$.
4. Insert the compact projected cores into `Bc_gamma/scripts/bc_ps_complete_analysis.py`. The script uses the diagonal calibration $M_1^2 = M_2^2 = 2M^2$, with $M^2 = \{7, 8, 9\}\,\mathrm{GeV}^2$ and $s_0 = \{53, 54, 55\}\,\mathrm{GeV}^2$.
5. Extract f_1 and f_2 from the diagonalized two-point perturbative moments and then sample $m_b, m_c, M^2, s_0, \theta$, and f_{B_c} in the Monte Carlo.
6. Rotate the basis amplitudes with $\widehat{\Pi}_{6743} = \sin\theta\,\widehat{\Pi}_A + \cos\theta\,\widehat{\Pi}_B$ and $\widehat{\Pi}_{6750} = \cos\theta\,\widehat{\Pi}_A - \sin\theta\,\widehat{\Pi}_B$.
7. Convert the pseudoscalar final-state coupling to the width using

$$\Gamma(1^+ \to 0^-\gamma) = \frac{\alpha_{\mathrm{em}}}{3} g^2 \left(\frac{m_i^2 - m_{B_c}^2}{2m_i}\right)^3.$$

At this stage the $B_c\gamma$ entries are the controlled hard-photon baseline with a conservative G^2 screen. The $B_c^*\gamma$ entries are leading perturbative vector baselines with a completed denominator-level G^2/contact inventory. The contact audit shows no ordinary two-channel contact contribution; the remaining explicit G^2 task is the numerical double-dispersion integration of the 538 all-positive denominator terms, with the crossed-contact rows kept separately as a consistency check.

11 Next Step

The double-dispersion variables satisfy

$$P^2 = p^2 + 2p \cdot q, \qquad pq = \frac{P^2 - p^2}{2}.$$

For $B_c\gamma$, the structure is close to the strange-sector E1 case, but with two heavy masses. For $B_c^*\gamma$, the additional $1/p_2$ and pq/p_2 structures mean that the double dispersion should be written explicitly before numerical Borel analysis. A diagonal single-variable replacement can be useful as a calibration check, but it should not be the final derivation for the vector final state.

Double-dispersion denominator

For the photon emitted from the charm line, the three denominators are

$$D_1 = (k+q)^2 - m_c^2, \qquad D_2 = k^2 - m_c^2, \qquad D_3 = (k-p)^2 - m_b^2.$$

With Feynman parameters x, y, z, $x + y + z = 1$, the combined denominator is

$$xD_1 + yD_2 + zD_3 = \ell^2 - \Delta_c,$$

where

$$\Delta_c = (1-z)m_c^2 + zm_b^2 - zy\,p^2 - xz\,P^2.$$

For photon emission from the $\bar{b}$ line, using the momentum routing in the FeynCalc scripts,

$$D_1 = k^2 - m_c^2, \qquad D_2 = (k-p)^2 - m_b^2, \qquad D_3 = (k-p+q)^2 - m_b^2,$$

and the analogous reduction gives, with $x + y + z = 1$,

$$\Delta_{\bar{b}} = xm_c^2 + (1-x)m_b^2 - x(1-x)p^2 + xz(P^2 - p^2).$$

Equivalently one may relabel the Feynman parameters after choosing a different loop momentum. These forms make clear why the true radiative problem is a double-dispersion problem in p^2 and P^2, not simply the two-point mixing correlator with an extra charge factor.

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